

Full Stack Development with MERN

Project Documentation

1. Introduction

Project Title: OrderOnTheGo

Team Members:

- **Rangisetty Priyanka:** (Team Leader) Backend Development, Schemas & Controllers
 - **Raavi Aruna :** Backend Development, Rotes & API Integration
 - **Raavi Lakshmi Sowjanya:** Frontend Development, UI Design & Implementation
 - **R.Bhavani:** Frontend Development, Authentication, Cart & Order Logic
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2. Project Overview

Purpose

OrderOnTheGo is a food ordering platform built with the **MERN stack**. The goal is to connect users with local restaurants through an easy-to-use, secure platform that allows browsing, ordering, and managing orders. Restaurant owners can manage their menu, and admins can promote restaurants and oversee the platform's operation.

Features

- User authentication (sign up, login, JWT token)
 - Browse restaurants, products, and categories
 - Cart management (add/remove items)
 - Order placement and tracking
 - Restaurant management (add, update, delete products)
 - Admin dashboard for restaurant promotions
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3. Architecture

Frontend

The frontend is built using **React.js**. It includes:

- **React Router** for dynamic page routing.
- **Redux** (optional) for state management.
- Custom components for user interface design, including reusable components like the header, footer, cart, and product cards.

Backend

The backend uses **Node.js** and **Express.js** to handle:

- API routes for user authentication, cart management, and order processing.
- Role-based access control (user, restaurant, admin).
- JWT token authentication for secure communication.

Database

The database uses **MongoDB** for storing:

- **Users** (authentication data)
 - **Restaurants** (details, products, orders)
 - **Orders** (order status, payment tracking)
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4. Setup Instructions

Prerequisites

- **Node.js**: Version 14.x or higher
- **MongoDB**: Installed locally or using MongoDB Atlas
- **npm**: Node package manager

Installation

1. **Clone the repository:**
git clone https://github.com/your-repo/OrderOnTheGo.git
2. **Install Dependencies:**
 - For **Frontend** (React):
 - cd client
 - npm install
 - For **Backend** (Node.js, Express):
 - cd server
 - npm install
3. **Set up environment variables:**
 - Create .env files in both client and server directories:
 - **Frontend**: Set the API base URL (REACT_APP_API_URL)
 - **Backend**: Set MongoDB URI (MONGO_URI), JWT secret (JWT_SECRET)
4. **Start the Application:**
 - For **Frontend**:
 - cd client
 - npm start

- For **Backend**:
 - cd server
 - npm start
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5. Folder Structure

Client

- client/: React.js frontend folder.
 - src/
 - components/: Reusable UI components (e.g., Header, Footer)
 - pages/: React components for different routes (e.g., Home, Cart, Orders)
 - redux/: Optional state management files (if Redux is used)

Server

- server/: Node.js backend folder.
 - controllers/: Logic to handle API requests.
 - models/: MongoDB schema models.
 - routes/: API routes (e.g., user, restaurant, order).
 - middleware/: Custom middleware (e.g., authentication, error handling).
 - config/: Environment variables and database connection.
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6. Running the Application

To start the application locally, follow these commands:

- **Frontend:**
 - cd client
 - npm start
 - **Backend:**
 - cd server
 - npm start
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7. API Documentation

User Endpoints

- **POST /api /register**: Register a new user
- **POST /api /login**: Log in and receive JWT token

- **GET /api /profile:** Get user details (Protected route)

Restaurant Endpoints

- **POST /api/restaurants/login:** Restaurant login
- **GET /api/restaurants/:id/products:** Get products for a specific restaurant
- **POST /api/restaurants/:id/products:** Add a product

Order Endpoints

- **POST /api/orders:** Create a new order
- **GET /api/orders/:id:** Get order details

Admin Endpoints

- **POST /api/admin/promote/:id:** Promote restaurant to homepage
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8. Authentication

Authentication in *OrderOnTheGo* is handled using **bcrypt** for secure password hashing and **React Context API** for client-side authentication state management.

- **Password Handling:**
 - User passwords are hashed using bcrypt before being stored in the database.
 - During login, the hashed password is compared with the entered password securely.
 - **Session Management:**
 - Once a user logs in, their information (like user ID, role, etc.) is stored in React's **Auth Context**.
 - This context is used across the app to control access to protected routes and display appropriate UI components (e.g., user vs. restaurant header).
 - **Role-Based Access:**
 - React Context tracks user roles (user, restaurant, admin) to ensure each role sees the correct dashboard and has access to relevant features.
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9. User Interface

Screenshots showcasing key UI elements:

- **Homepage:** Displays restaurant categories and popular restaurants.
 - **Cart Page:** Lists added items with the option to update and checkout.
 - **Restaurant Dashboard:** Allows restaurant owners to manage their products.
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10. Testing

Testing Strategy

- **Unit Tests:** Tests for backend API routes, models, and controllers.
- **Integration Tests:** Full-stack testing (frontend to backend communication).
- **Manual Testing:** Functional testing of user flows such as login, cart updates, and order placement.

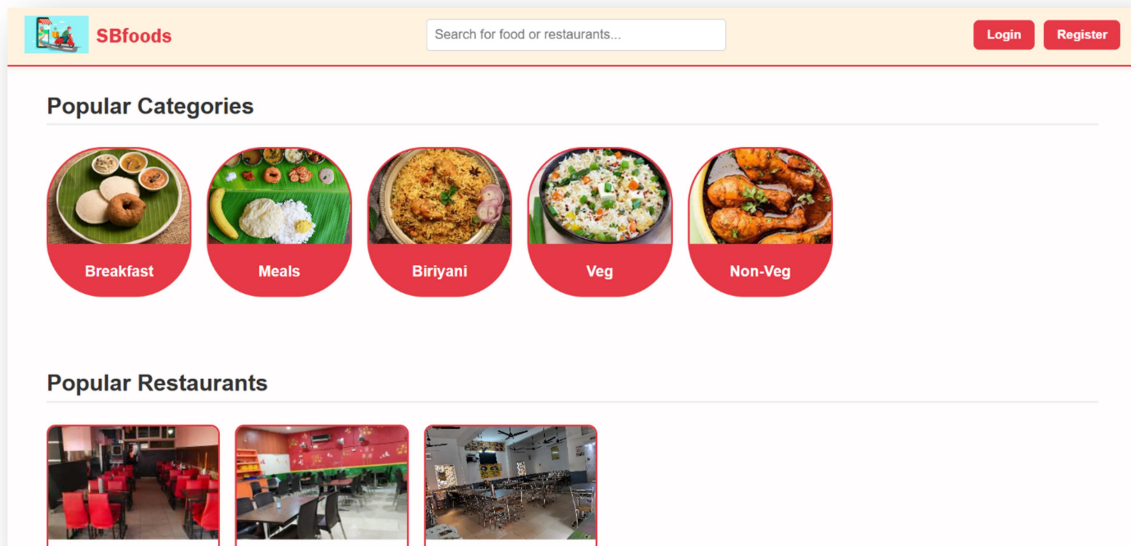
Tools Used

- **Jest** for unit testing
- **Supertest** for API testing

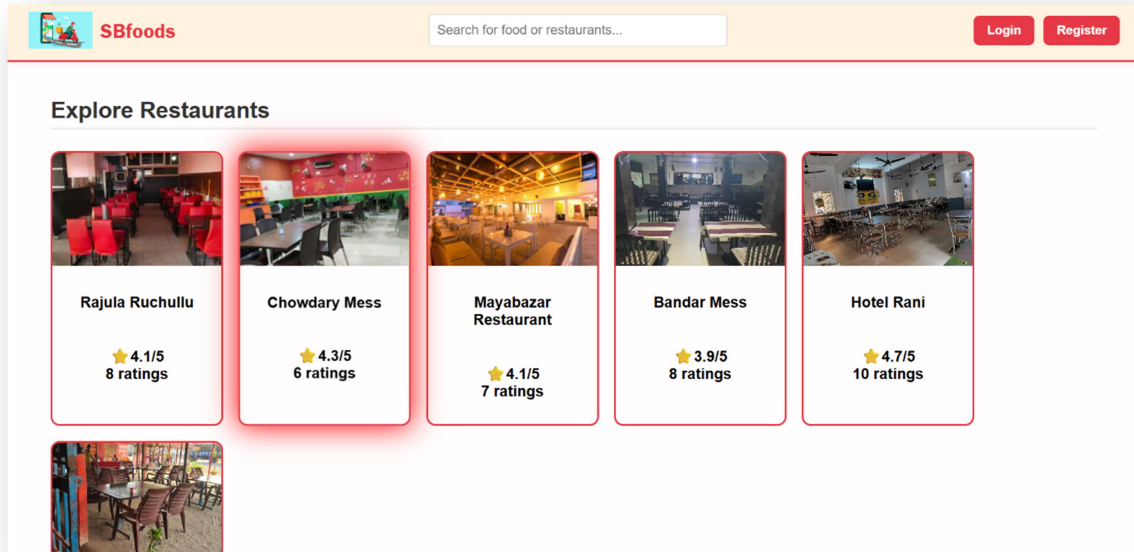
11. Screenshots

- **Screenshots:**

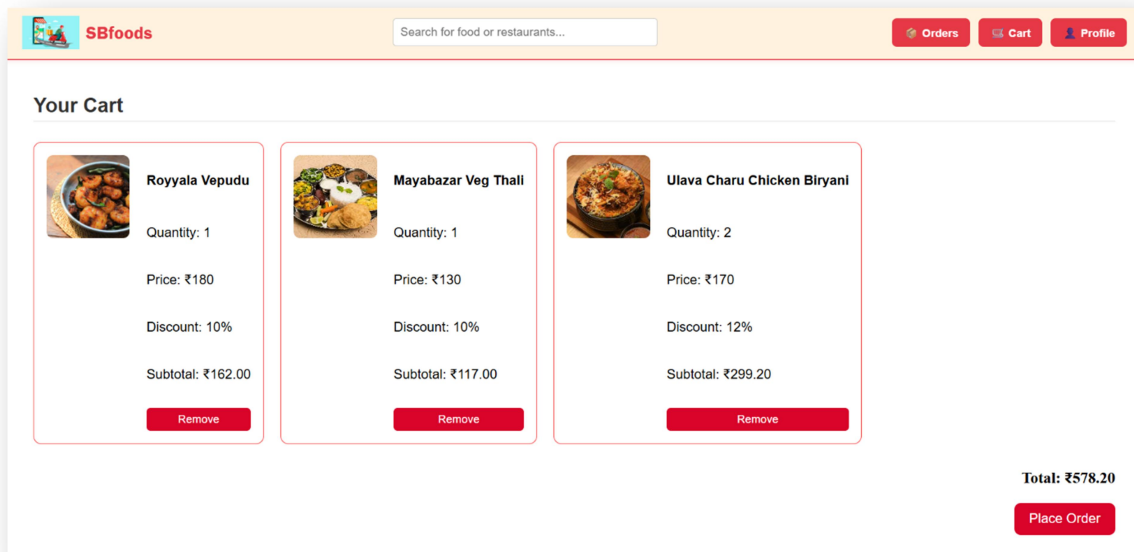
Homepage:



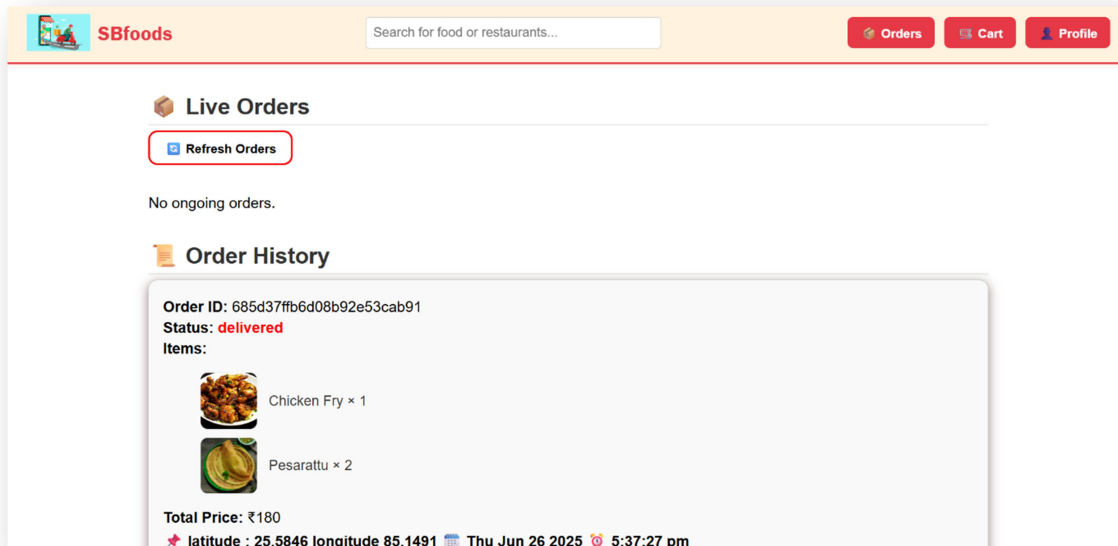
Restaurants:



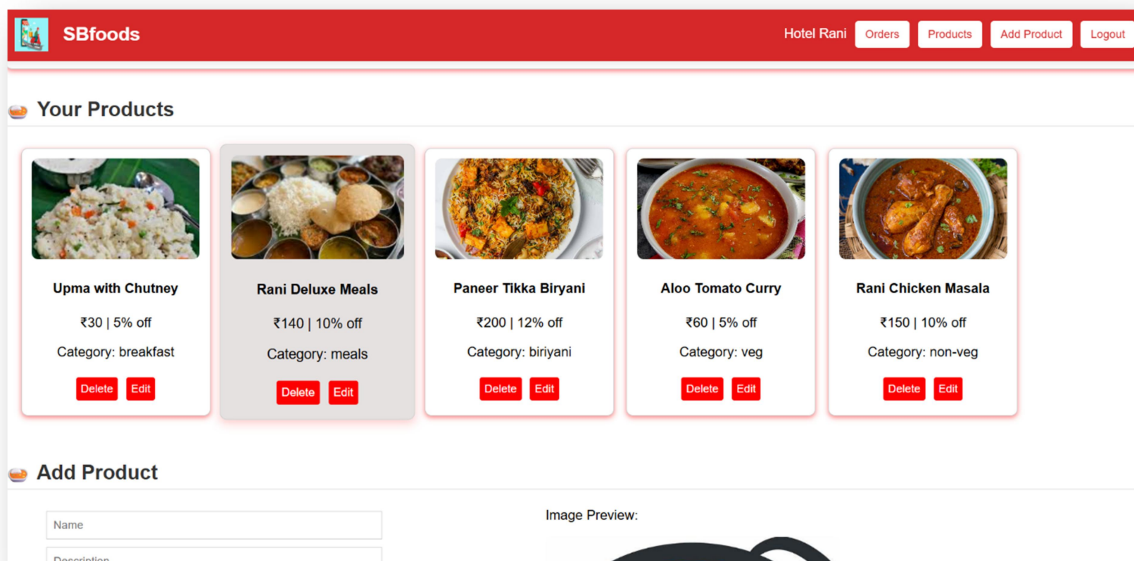
Cart:



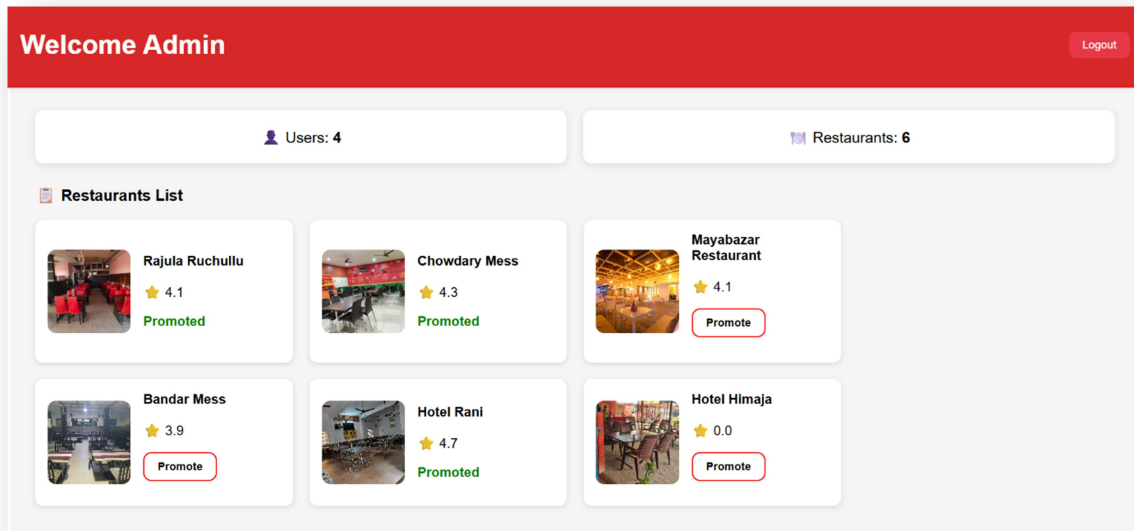
Orders:



Restaurant Dashboard:



Admin Dashboard:



12. Known Issues

- **Payment Integration:** Not implemented yet, placeholder functionality.
- **Mobile App:** Currently no mobile app version, only web-based.

13. Future Enhancements

- **Payment Gateway:** Integrate with a real payment provider (e.g., Stripe, Razorpay).
- **Mobile App:** Develop a mobile version using **React Native**.
- **Notifications:** Implement order status notifications for users and restaurants.
- **Analytics Dashboard:** Add real-time analytics for restaurants and admins.